पंचायती राज मंत्रालय
Ministry of Panchayati Raj

एनआईसी National Informatics Centre

IIT Tirupati Navavishkār I-Hub Foundation

भारतीय प्रौद्योगिकी संस्थान तिरुपति
TIRUPATI

National AI/ML Hackathon

Final Round Presentation

Di2S2-Net

Dino Drone Semantic Segmentation Network

End-to-end, fully-automated feature extraction from drone orthomosaics :- a satellite-pretrained DINOv3 ViT-L/16 + HRDecoder turning raw imagery into a GIS-ready GeoPackage of village assets.

Team Id Number: **AIML-F-028**
Organization / Institution: **EarthSense Labs**
Problem Statement: **PS-1 — AI-Based Feature Extraction from Drone Images**
Presented by: **Nirdesh Kumar Sharma** · **Surampudi Samvedya** · **Gaurav Shende**

Proposed Solution

An end-to-end, fully-automated pipeline that turns a raw drone orthomosaic into a **GIS-ready GeoPackage** of village assets , **no manual digitisation** , built on a satellite foundation model rather than ImageNet.

WHAT WE BUILT

One automated pipeline, raw ortho → GeoPackage. Auto CRS-reproject and COG-tile a multi-GB orthomosaic, segment every tile, then memory-safely stitch and vectorise into per-class polygons & road centre-lines , **zero manual digitisation**. Shipped as a CLI batch tool **and** a no-code web **portal** for field staff.

Satellite foundation model + two-pass HRDecoder. A **DINOv3 ViT-L/16** encoder pretrained on **493 M satellite images** brings domain-matched features, so only a light head trains on our 10 labelled villages. Its **HRDecoder** fuses a global low-res pass with native-res 256^2 windows , capturing huge-area **and** thread-thin assets from the **same** tile.

7 TARGET CLASSES

% of labelled pixels

- Background 71.7%
- Built-Up 17.2%
- Road 5.4%
- Water Body 5.1%
- Utility 0.6%
- Bridge ~0%
- Railway 0%

In the labelled corpus, **Railway has no ground truth anywhere** and **Bridge is vanishingly rare ,** just 12 of 4,513 tiles (0.004% of pixels). Both score IoU≈0 by **data coverage**, not model failure, so every result is reported **with** and **without** them.

FROM TILE TO MAP

real input → prediction

Orthomosaic

One village · multi-GB GeoTIFF at **~3 cm** GSD

↓ tile · 1024^2 · 128-px overlap

Tiles

34,708 tiles across 20 villages

↓ batched inference

Di2S2-Net

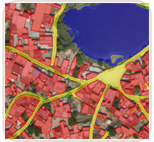
DINOv3 ViT-L/16 encoder + HRDecoder → 7-class logits

↓ argmax · per tile

Predicted tiles

7-class mask · one per tile

↓ stitch + vectorize



Combined output

Stitched, georeferenced GeoPackage · QGIS-ready

Powered by: Geospatial Intelligence and Applications Laboratory, IITTNiF

Key Challenges → Our Approach

Five hard problems shaped every design decision in Di2S2-Net — each addressed **and** validated on real data.

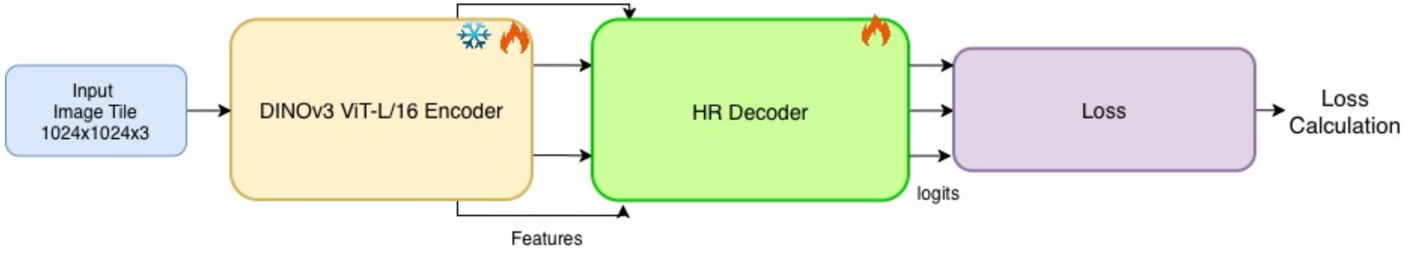
#	THE CHALLENGE	OUR SOLUTION	EVIDENCE
1	**One tile, two scales** huge-area + thread-thin features at 3 cm GSD	**HRDecoder two-pass fusion** — global low-res context + native-res 256^2 sliding-window detail, count-map fused.	Water .96 · Built .95 Road .88 · Utility .77
2	**Scarce, clustered labels** 10 / 20 villages, only 2 states	**DINOv3 ViT-L/16 (SAT-493M)** foundation — fine-tune top 12 of 24 blocks → a light ~3 M-param head.	0.905 in-sample 0.73 unseen villages
3	**Severe class imbalance** background 71.7% vs Utility 0.57%	**Combined loss** 0.5·CE + 0.3·Dice + 0.2·Edge, multi-scale supervision (Fuse / LR / HR).	minorities recovered Utility .77 · Road .88
4	**Messy raw geodata** mixed CRS · ECW + TIF · silent dupes	**Canonical-key de-dup** + **auto-UTM COG** conversion (DEFLATE, 512-px, centroid zone).	20 villages, no leakage metre-accurate output
5	**Rasters exceed GPU / RAM** KURTU village = 10,523 tiles	**Constant-memory windowed stitching** + COG windowed reads + FP16 inference.	34,708 tiles · 1×24 GB ~7 tiles / s

ALSO ENGINEERED

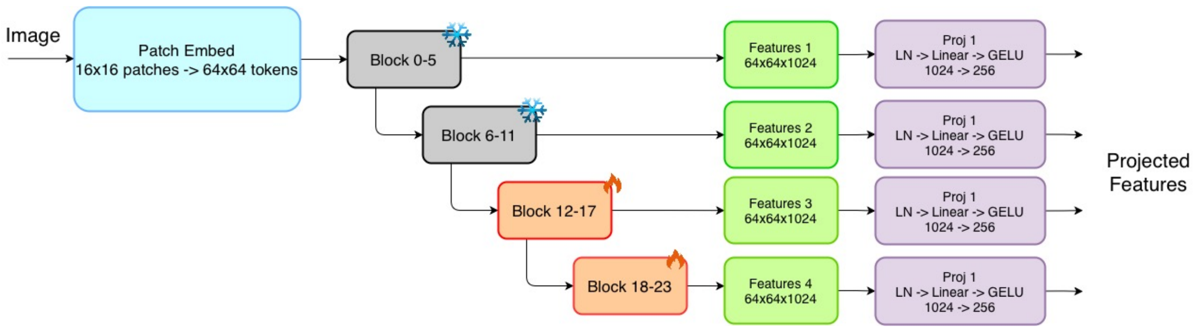
GIS-ready GeoPackage(per-class polygons + road skeletons) ·**cheap transfer**(freeze encoder, swap ~3 M head) · **fault-tolerant batch** + checkpoint-architecture guard · the **Portal** (browse · run · compare, no CLI) · **honest metrics** reported with/without empty classes (+0.258 mIoU).

System Architecture / Workflow

MODEL SPINE 1024² tile → encoder → HRDecoder → multi-scale loss



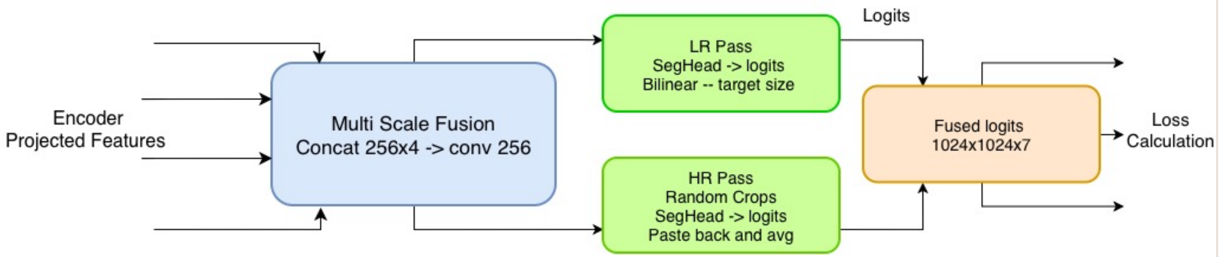
ENCODER DINOv3 ViT-L/16 · SAT-493M



24 blocks · 1024-dim · patch 16→ 64×64 tokens; taps [5, 11, 17, 23] → 4 scales.

Each scale LN→Linear(1024→256)→GELU; first **12 blocks frozen**, 12–23 fine-tuned at **0.1× LR**.

DECODER HRDecoder · two-pass fusion



Fusion concat 4×256 → conv 256.**LR pass** = full-image context; **HR pass**= 256² crops for detail.

Train: x random crops; test: deterministic sliding window + count-map → **fused 1024×1024×7**.

Productisation — the same binaries are wrapped by a **FastAPI + TiTiler** backend and a **React + MapLibre GL** client a modular decoder registry (HRDecoder · UPerNet · SegFormer · SkipDecoder) lets the portal sniff a checkpoint's architecture and refuse incompatible ones.

HRDecoder — Two-Pass Fusion

One pass on the coarse feature grid **blurs** thin features; a full high-res pass is too costly. **HRDecoder runs both , a global low-res pass for context and a local high-res pass for detail >> then fuses them.**

INPUT TILE

$1024^2 \times 3$

→

LR · GLOBAL PASS

One SegHead over the **whole tile** at the coarse 64^2 feature grid → upsampled to 1024^2. Large areas & **context** are right, but **boundaries stay soft ,** see the blocky edges.

HR · LOCAL PASS

256^2 windows decoded at **native resolution** & count-map averaged. Thin **roads, utility** and building outlines come back **crisp**.

⊕

FUSED OUTPUT

$1024^2 \times 7$

TRAIN — RANDOMISED

2 random 256^2 crops (jittered ×0.75–1.25, snapped to ÷8), each SegHead'd and **averaged with the LR base** :- the randomness doubles as **regularisation**.

TEST — DETERMINISTIC

A **sliding window** of 256^2 crops tiles the image; logits accumulate in a **count-map** and are averaged → **seamless, reproducible** full-res output.

Net effect— large-area recall from the LR pass **and** thin-feature precision from the HR pass, fused at **native 1024^2** at constant memory: Road IoU up to **0.88** with crisp, geo-referenced boundaries.

Implementation & Technical Approach

DATA ENGINEERING — DECISION → WHY

Dedup by canonical key (strips ext / _3857 / case) keeping valid>corrupt, tif > ecw → no train/test leakage.

COG— auto-CRS; EPSG:4326 reprojected to centroid **UTM zone** ; DEFLATE, 512-px tiles → metre-accurate areas + windowed access.

Tiling 1024^2, 128-px overlap (step 896); drop tiles >50% no data.

Labels — shape files → uint8 via R-tree + (class,crs) re project cache; points buffered **3 m**.

TRAINING RECIPE

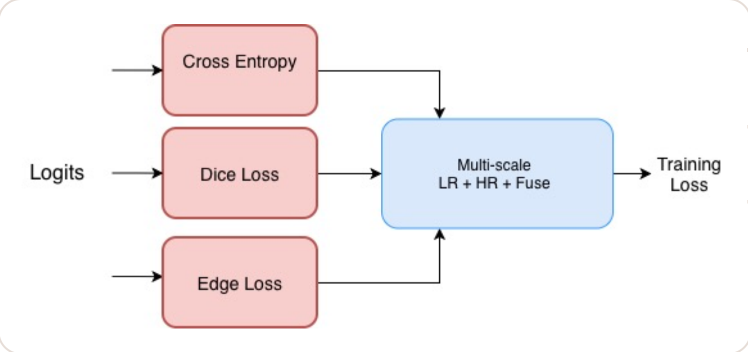
train_full.py

Framework	Lightning · FP16 (16-mixed)
Tile / batch / accum	1024^2 · 4 · 2 (eff. 8)
Optimizer · clip	AdamW wd 1e-4 · 1.0
Base / encoder LR	1e-4 / 1e-5 (0.1×)
Schedule	cosine · 5-ep warmup
Epochs · best	50 · train/loss 0.0615
Hardware	1× 24 GB GPU

≈**300 M** params · ~**155 M** trainable · task head (proj + decoder) only ~**3 M** → freeze encoder to port a new region in hours.

LOSS DESIGN

CombinedLoss · multi-scale



TERM	W
Cross-Entropy classify	0.5
Dice imbalance	0.3
Sobel boundary consistency	**0.2**

Multi-scale supervision Total = 1.0·Fuse + 0.5·LR + 0.1·HR. Boundary precision comes from the **HR pass at native 1024^2** + Dice; the FP16-safe Sobel term is a low-weight consistency regulariser.

ENGINEERING FOR SCALE & ROBUSTNESS

FP16 autocast inference (~50% latency cut); GPU cache flushed every 50 batches → constant VRAM.

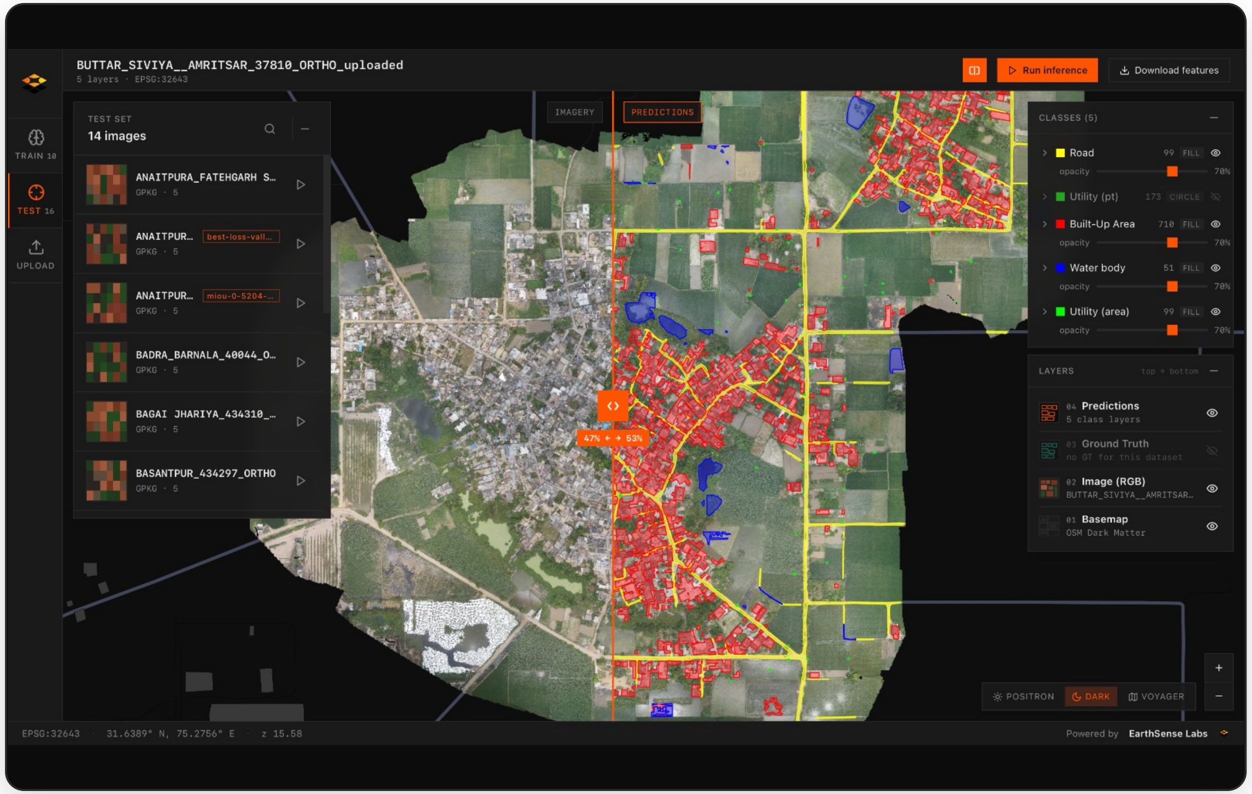
Windowed stitching (512^2 blocks, last-write-wins) rebuilds full rasters in **constant memory** — handles the 10,523-tile KURTU village.

Skip-resume orchestration +**checkpoint architecture guard** (sniffs decoder from state-dict, aborts on mismatch).

Vectorise → simplify + min-area; **road skeletons**, utility negative-buffer; multi-layer .gpkg + QGIS styles.

Results & Demonstration

LIVE PORTAL swipe compare · imagery ↔ predictions



Browse pre-computed results + **run live inference** (5 phases, streamed over SSE) on a new orthomosaic :- same binaries, so demo == submission.

Swipe imagery vs prediction · **multi-checkpoint** compare (name spaced, architecture-guarded) · one-click GeoPackage download.

0.905	0.647	~98.4%	0.73
mIoU · 5-class present	mIoU · 7-class w/ B+R	Overall accuracy	mIoU · unseen villages

PER-CLASS IOU

full train · in-sample

Water	0.956
Built-Up	0.949
Road	0.879
Utility	0.765

GENERALISATION

held-out villages

EXPERIMENT	7-CLS	5-CLS	OA
Full · 10 villages	.647	.905	.984
9:1 · NAGUL	.520	.729	.951
8:2 · TIMMOWAL+NAGUL	.522	.731	.935

~0.90 in-sample, **~0.73 on never-seen villages** from only 10 training villages; the 0.65 figure is purely the artefact of averaging in two zero-GT classes. Throughput **~681 tiles / 99 s**(≈7 tiles/s).

Portal offline? Watch the recorded walkthrough — drive.google.com/…/Di2S2-Net-demo

Innovation & Key Learnings

WHAT'S NOVEL

Satellite foundation model for SVAMITVA :- DINOv3 SAT-493M weights (not ImageNet) make 10 villages enough.

HRDecoder two-pass fusion solves the multi-scale curse; train-time random crops double as regularisation, test-time sliding-window is seamless.

Multi-component loss + multi-scale supervision (Fuse / LR / HR = 1.0 / 0.5 / 0.1).

District-scale engineering :- constant-memory stitch, FP16 inference, skip-resume, GIS-native GeoPackage with road skeletons.

Pipeline → product :- an architecture-aware portal that **wraps**, never re-implements, the pipeline.

KEY LEARNINGS — PROBLEM → FIX

The metric lied (–0.26 mIoU). Railway has zero GT and Bridge is vanishingly rare (12/4,513 tiles, 0.004% of pixels) → both unlearnable, pinned to IoU≈0, dragging 7-class to 0.65 vs true 0.905. Report both; head reserves IDs 5/6, so adding labels needs no re-architecture.

Memory wall at stitch (10 GB+ for KURTU) → windowed writes → constant memory.

FUTURE WORK

Annotate **Bridge, Railway & Utility :-** Utility is the hard class (IoU 0.38–0.47 held-out: small, sparse targets).

More villages → close the in-sample **0.90** vs held-out **0.73** gap; cheap via the ~3 M-param frozen-encoder transfer path.

Make the edge term **differentiable** (soft-probability Sobel).

Backbone upgrade :- DINOv3 ViT-7B / DINOv4 behind the same wrapper.

Multi-GPU + **Docker / CUDA** for district rollout.

Open-source only — PyTorch · Rasterio · GDAL · GeoPandas · scikit-image · FastAPI · MapLibre. No proprietary deps.

Q&A

Thank You

Di2S2-Net — Dino Drone Semantic Segmentation Network. Questions & discussion welcome.

DINOv3 ViT-L/16 · SAT-493M

HRDecoder · two-pass

PyTorch Lightning · FP16

Rasterio · GDAL · GeoPandas

FastAPI · TiTiler

React · MapLibre GL